

Lecture 13

8051 microcontroller
ROM, RAM, Decoder, I/O Devices, Peripheral Devices,
etc to make a MC
Also interrupt Controller, programmable I/O devices,
DMA controllers, To improve the capability & performance
MP based system $\rightarrow$ more flexibility,
more instruction set,

ROM, RAM, Parallel/serial I/O, Counter
CU circuit, all inbuilt, Computer on
chip,

Data transfer will be faster, than
internal interfacing as compared
To on chip, no external wiring

MP	MC
ALU, CU (CU/Timing ckt) different registers, interrupt ckt.	Contains, MP, memory (ROM + RAM) I/O interfacing ckt, Peripheral devices such as, serial I/O, (ADC, DAC), timer.
- many instructions to transfer data b/w memory & CPU	1-2 instructions to transfer data b/w mem & CPU
1-2 bit handling instructions more access time	many bit handling (To handle flag registers) less access time
more h/w, more flexible single memory map for data and code.	less H/w, less flexible. separate memory map for data & code
less no of pins are multif.	more pin are multifunctional.

features of 8051

• 4kB on chip Memory (ROM / EPROM) Program

• 128 bytes on chip data memory

• 4 Register bank

• 64kB each program and external RAM addressability.

• 1H S instruction cycle,
12 MHz Crystal

- 32 bidirectional I/O lines

organized as 4 - 8 bit ports.
Multiple modes high speed

programmable serial data (UART)
- 16 bit timer / counters.

- Direct Byte + Bit addressability

Accumulator

- general Register To accumulate Result.

Can hold an 8-bit (1 byte) Value.

B register - 8 bit data storage,
multiply AB, Divide AB.

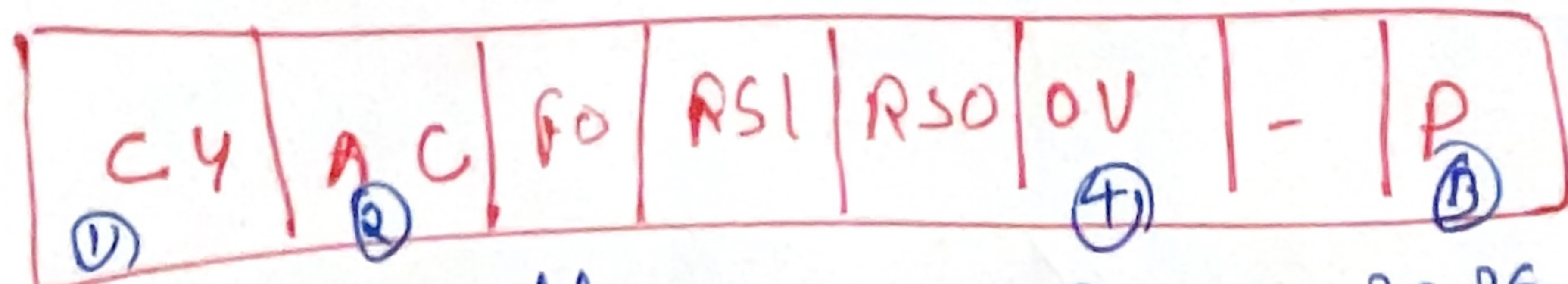
PSW (Program status word)

8 bit, 6 bits used, 2 bits

4 bit
Condition

a bit
To select
Register
bank

MSB



• No zero flag
• zero, sign flag

④ OV in 8086
Not in 85

40 available for user

RS1, RS0 - To select Register bank

RS0, RS1

00 → Bank 0 (00H - 07H)
01 → Bank 1 (08H - 0FH)
10 → Bank 2 (10H - 17H)
11 → Bank 3 (18H - 1FH)

8 Reg's each bank

Stack Pointer (8 bit)

16 bit for RS, RS

PUSH - SP decrement (call subroutine)
POP - SP increment.

07H after Reset SP

When we call subroutine, we
Push the next instructions in the
stack. (address pushed)

Data Pointer (DPTR)

• high byte (DPH) & low Byte (DPL)
• 16/8 bit 2 Regs.

PC (Program counter) - 2 byte (0000h)

a byte address which tells
8051 where the next instruction
To execute is found in memory

PC - starts at 0000h
, incremented each time ins-
truction is executed -

JUMP-instruction I can alter PC.

PORTS 0 to 3.

P0, P1, P2, P3 - latch with the help of SR latches.

Writing a 1 to a bit of port SFR (P0, P1, P2 or P3) causes the O/P pin to switch ~~off~~ high.

When used as I/P the external state of a port pin will be held in the Port SFR.

SR latch to store values of the port.

Serial Data Buffer

2 separate registers, a transmit & a receive buffer.

Sending To SBUF - transmit buffers
& loading from SBUF - receive

Timer Register,

Register Pairs (TH0, TL0)
(TH1, TL1) @ 16 bit each

16 bit Counting registers,
Control Registers for 8051

IP (interrupt ptr, Interruptable),
IE, TMOD, TCON, SCON,

PCON

(Timer mode, Counter)

52 PCON

Register banks, 8 "R" Registers

8 Registers, (R0 - R7)

- Data move, transfer, manipulation -

PSEN Program store enable
4 Control signals, that enables
Program Control (memory)
connected to an EPROM, output enable
OE(Pin) To permit reading of program
bytes

PSEN - low - fetch → mem knows to read instruction
execute - high -

internal memory is divided into
register banks, (00H - 1FH)
bit addressable RAM (20H -
2FH)

③ general purpose RAM
(30H - 7FH)

④ special function

(80H - FFH)

Registers in RAM

32 registers - 4 banks.

PSW's - 2 bit which bank

bit addressable RAM

16 Registers

(20H - 2FH)

(bit addressable).

general purpose RAM

(30H) (hex) → decimal.

32 16 8 4 2 1
00110000

481

2FH

00101111

32 15

47

30H

48 32 16 8 4 2 1

01001111

ALE.

for demultiplexing the address + data bus.

Data Bus and lower byte of address bus

are multiplexed

EA, (+5V or 0V) permanently high or low.

- high - 8051 will execute program via ROM

- low - Program executes from external memory (bit addressable).

RST - master reset

high for at least 2 machine cycles,

8051 internal registers are loaded
with appropriate values for an orderly
system start up.

- on chip oscillator

12 MHz frequency

Program → (runtime (execution time) it
will go inside RAM will eat RAM space

How many programs are in RAM.

HP - Related limited memory

HP → Different spaces, At Runtime

in HC → Controller, ROM & RAM

are inbuilt

Different Prog + Data Memory.

Bit addressable.

8051 contains 210-bit-addressable locations.

128 are in 20H through 2FH and rest in special function registers - individually accessing bits through software is painful.

Bit can be set, cleared

(set, cleared, AND, OR)
We haven't talked about bit inside memory, we talked about byte.

210 - 128 bit 20H - 2FH
in MP, if I want to achieve same effect, then,
~~Read~~

If I want to modify any bit in a MP, then

- 1 Read data
- 2 Put in accumulator.
- 3 Modify
- 4 Memory write

But in HC - we can give direct instruction to modify any bit in memory.
I/O Ports are also bit addressable.
Register banks.

last 32 locations of internal memory
8 Registers, (R0 to R7) are at address

00H - 07H } short and fast instructions
when we use BC, DE, HL } fast cause
inside memory.

Special function Regs.

Part of on-chip RAM.

each Reg has also an address,

Many Registers in PS.

there are 21 (SFR's)

special function registers

(80H to FFH)

External Memory.

64k External Code Memory

64k External Data Memory

ROM + RAM } extension

Peripheral IC
Interface IC

can also be used
Memory mapped I/O.

Port 0 - Address + data
will be multiplexed.

64k B

2^{16}

16 address lines

A0 - A7 - Port 1

A8 - A15 - Port 2

Port 1 is multiplexed

P1 & P2 will receive address

Data with Port 1

ld-

8 Registers, 4 Banks

00 to 1F

↑ Bank 3

Bank 0

20 - 2F (Other Registers)

Bit addressable Mem

30 - 7F.

20 - 2F (16 Regs) (bit

00 - 1F - 32 Registers 4 Banks.

20 - 2F - Bit addressable Mem

30 - 7F - general Purpose RAM

80 - FF - Special Function Registers -

RS1 RS0 | User Control Values in PSW.

0 0
0 1
1 0
1 1

128 bits - 00 to 7F H

each bit Variable Can be set or clear.

set B → Bit '1

CLR Bit → 0

general Purpose Mem - 80 - byte

00 - 2F

30 to 7F H by using direct / indirect

Addressing Mode.

$\overline{PSEN}$ → Program ~~sa~~ To be used for RAM or ROM.

Port 0 - Address/ Data Multiplexed

Pin 1 to Pin 8 - P
input/output

Pin 9 Reset
(Active high)

Pin 10 - Pin 17 (Port 3)
I/O

INT0: Pin 12 } External
INT1: 13 } interrupt.

Pin 20 gnd

21 to 28 Port 2.

$\overline{PSEN}$ - 29 (ROM/RAM) - Instruction

$\overline{ALE}$ - 30

latch lower 8 bits -

32 to 39 - I/O

Addressing Modes

Immediate - Address directly inside instruction

Register -

Direct -

Register indirect -

Indexed -

Assembler $\rightarrow$ (Expression.)

$\downarrow$
Data

~~MOV~~

B MOV DPTR, #4521H

DPL - 21

DPH - 45

Q Register addressing -

ADD A, R5

ADD

MOV R7, PPL

Register to Register Transfer

is Not allowed -

MOV R7, DPL

Direct Addressing

MOV R0, 40H

- SFR (Special Function Register)

MOV 0F0H, #55

MOV A, #55R

SFR 00H & FFH

Special Function Registers

There will be some empty space as well

- But Not allocated to

Programmer.

Register indirect -
Register to hold address

MOV A, @R0

MOV @R1, 35H

Value inside R1 is address, store 35 there

@ Register's Value as address
Indexed Addressing

- Uses a base Register -

PC or Data Pointer, and an offset, the accumulator

later -

PC

or Data Pointer + Accumulator

To find effective address

for a Jump /

MOVC

DPTR/PC

I/O Ports:

4 8 bit I/O ports P₀, P₁, P₂, P₃

each uses 8 Pins.

Reset → Ports become O/P, ready

To be used as inputs ports.

Port 0 - AD₀-AD₇ - Address Data

if external memory.

low 8 bit address latched.

P₀ & P₂ are used as simple I/O.

(AD₀-AD₇)

if external Memory.

ALE = 0 (data)

ALE = 1 (address).

No external Memory -

P₀ & P₂ → simple I/O

if external memory

P₀, P₂ will take address

& data will be only there at

P₀.

HP - all Memory outside

HC - we can add extra memory

bus - address, latch, data lines

Counters & Timers

2 16 bit counters - T₀ & T₁

- To Count frequency of a pulse train

or precise internal time delays b/w computer actions.

To count internal clock (timer) > To count external pulses.

a) interval timing

b) event timing

c) Baud Rate gen

for the built in serial port.

2 Timer / Counter

- Timer overflow flag.

- flag synchronizes

TMOD Timer Mode Register

TMOD Timer Mode Register

2 groups of 4 bits, that set the operating mode for timer 0 & T₁.

M1	M0	Mode	
0	0	0	13 bit timer mode
0	1	1	8 bit timer/counter
1	0	2	16 bit timer mode
1	1	3	TRX, TLX, 8 bit auto reload
			split timer mode.

Auto Reload

TRX holds a value which is

To be Reloaded TLX each time it overflows

11 - TRH, T_L can be

used as diff timers.